ETL Process Using Power BI (OnlineRetail Dataset)

1. Extract (Data Collection)

- Open **Power BI Desktop**
- Import dataset using **Get Data → Text/CSV**
- Select **OnlineRetail.csv** (or any csv file of your choice) file
- Click on **Transform**

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2. Transform (Data Cleaning & Processing)

For now do not rename any column.

- Change data types of columns: (Right Click on the columns>change type)
  - InvoiceNo → Text
  - StockCode → Text
  - Description → Text
  - Quantity → Whole Number
  - InvoiceDate → Date/Time
  - UnitPrice → Decimal Number
  - CustomerID → Whole Number
  - Country → Text
- Remove rows with **null CustomerID** (Click on CustomerID column>top pane> Remove Rows>dropdown>Remove Null Rows)
- Filter **Quantity > 0** to remove returned items (Click on Quanity Dropdown>filter>numerical>greater than 0)
- Created a derived column **TotalAmount** using: (on top header>add column>custom column>put name as TotalAmount>add columns below> qty*unitprice)
  - `Quantity × UnitPrice`

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3. Load (Data Loading)

- Apply transformations using **Close & Apply**

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4. Visualization (Data Analysis)

- Create **Bar Chart** to show total sales by country:

Select Bar chart

In the rightmost pane select your csv>click on country and total amount

Drag Country in x axis, Total amt in y axis

If chart is too sparse or not visually appealing apply filter>drag country in filter select top N filter in dropdown>write 10>In by value drop TotalAmount Column

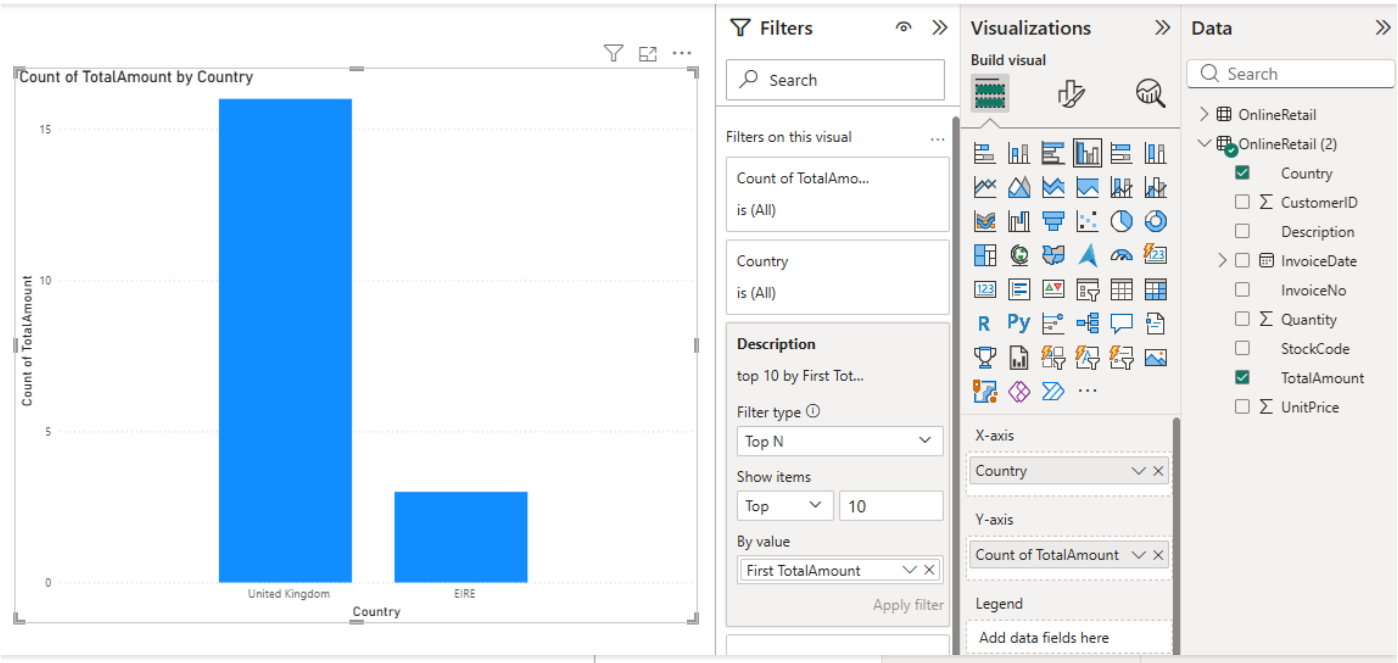
- Create **Pie Chart** for top contributing countries

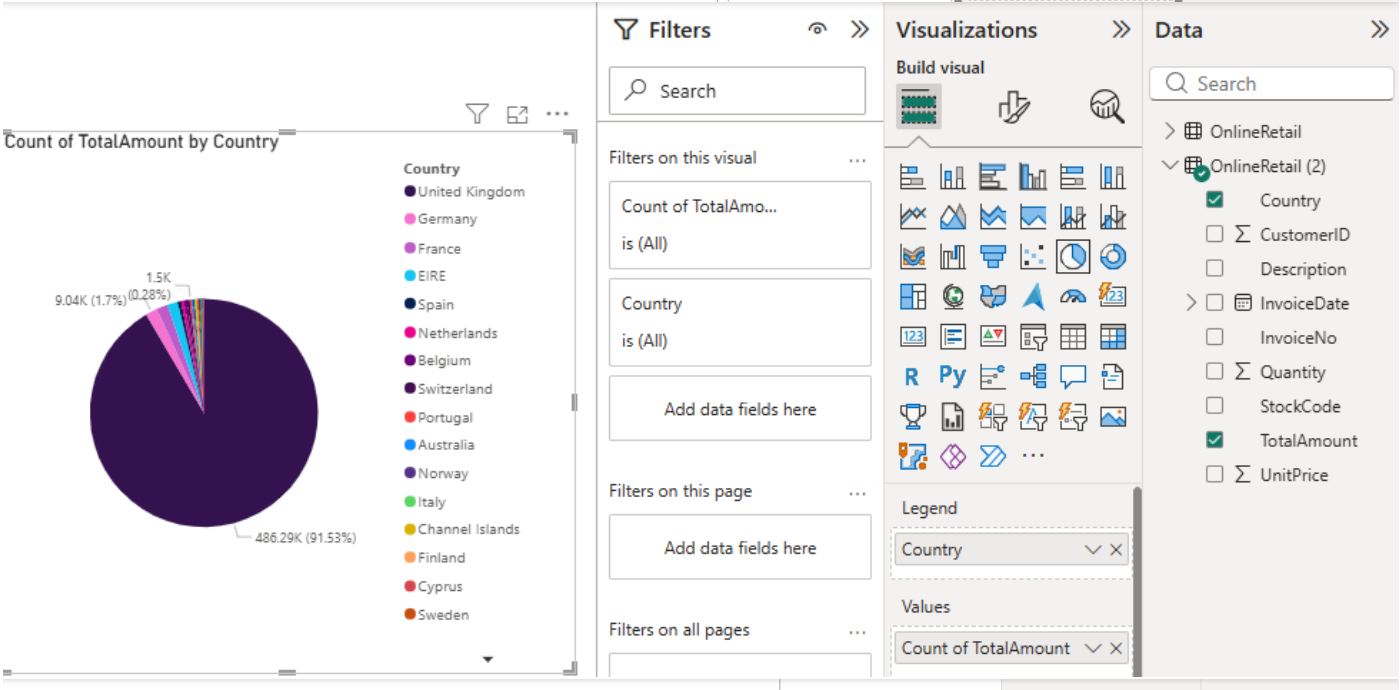
Select Pie Chart> select your csv>take country and total amount>legend is country and values are totalamount

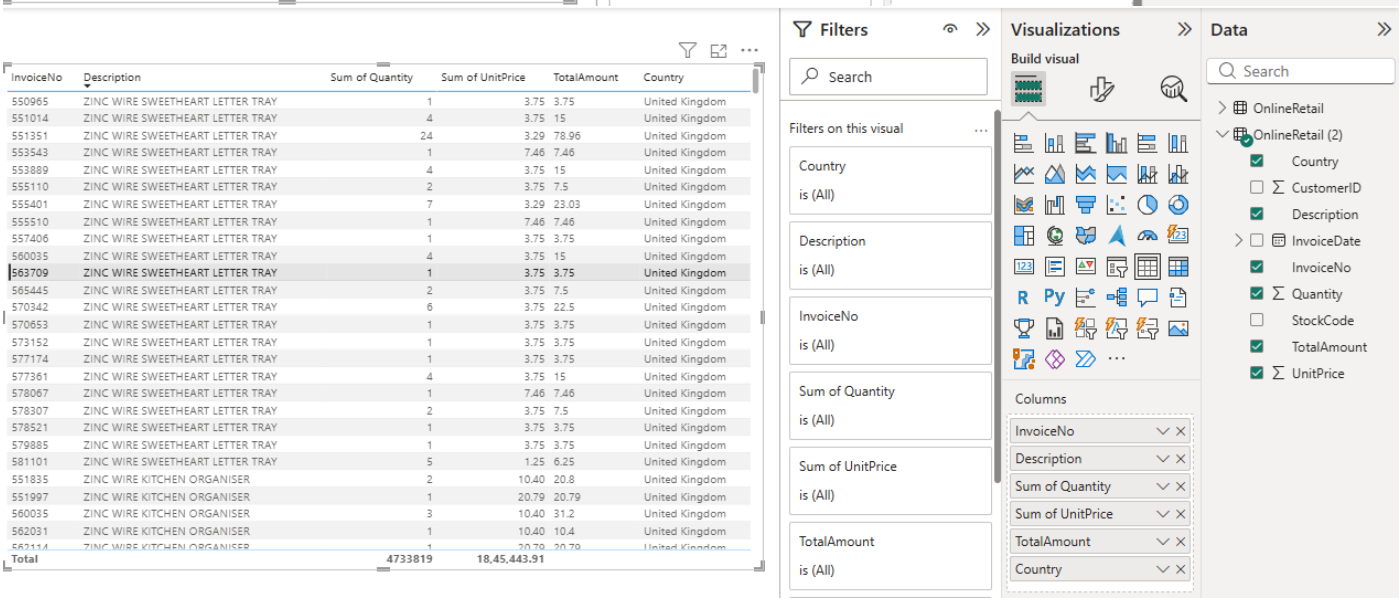
- Create **Table Visual** to display detailed sales records

Select Table>select your csv>take invoice no, description, quantity, unit price, total amount, country.

Table is ready. Filter can be applied to view by country by dropping country under filter and then select UK







InvoiceNo	Description	Sum of Quantity	Sum of UnitPrice	TotalAmount	Country
550965	ZINC WIRE SWEETHEART LETTER TRAY	1	3.75	3.75	United Kingdom
551014	ZINC WIRE SWEETHEART LETTER TRAY	4	3.75	15	United Kingdom
551351	ZINC WIRE SWEETHEART LETTER TRAY	24	3.29	78.96	United Kingdom
553543	ZINC WIRE SWEETHEART LETTER TRAY	1	7.46	7.46	United Kingdom
553889	ZINC WIRE SWEETHEART LETTER TRAY	4	3.75	15	United Kingdom
555110	ZINC WIRE SWEETHEART LETTER TRAY	2	3.75	7.5	United Kingdom
555401	ZINC WIRE SWEETHEART LETTER TRAY	7	3.29	23.03	United Kingdom
555510	ZINC WIRE SWEETHEART LETTER TRAY	1	7.46	7.46	United Kingdom
557406	ZINC WIRE SWEETHEART LETTER TRAY	1	3.75	3.75	United Kingdom
560035	ZINC WIRE SWEETHEART LETTER TRAY	4	3.75	15	United Kingdom
563709	ZINC WIRE SWEETHEART LETTER TRAY	1	3.75	3.75	United Kingdom
565445	ZINC WIRE SWEETHEART LETTER TRAY	2	3.75	7.5	United Kingdom
570342	ZINC WIRE SWEETHEART LETTER TRAY	6	3.75	22.5	United Kingdom
570653	ZINC WIRE SWEETHEART LETTER TRAY	1	3.75	3.75	United Kingdom
573152	ZINC WIRE SWEETHEART LETTER TRAY	1	3.75	3.75	United Kingdom
577174	ZINC WIRE SWEETHEART LETTER TRAY	1	3.75	3.75	United Kingdom
577361	ZINC WIRE SWEETHEART LETTER TRAY	4	3.75	15	United Kingdom
578067	ZINC WIRE SWEETHEART LETTER TRAY	1	7.46	7.46	United Kingdom
578307	ZINC WIRE SWEETHEART LETTER TRAY	2	3.75	7.5	United Kingdom
578521	ZINC WIRE SWEETHEART LETTER TRAY	1	3.75	3.75	United Kingdom
579885	ZINC WIRE SWEETHEART LETTER TRAY	1	3.75	3.75	United Kingdom
581101	ZINC WIRE SWEETHEART LETTER TRAY	5	1.25	6.25	United Kingdom
551835	ZINC WIRE KITCHEN ORGANISER	2	10.40	20.8	United Kingdom
551997	ZINC WIRE KITCHEN ORGANISER	1	20.79	20.79	United Kingdom
560035	ZINC WIRE KITCHEN ORGANISER	3	10.40	31.2	United Kingdom
562031	ZINC WIRE KITCHEN ORGANISER	1	10.40	10.4	United Kingdom
562114	ZINC WIRE KITCHEN ORGANISER	1	20.79	20.79	United Kingdom
Total		4733819	18,45,443.91		